Macroeconomics I — MPE Insper

Practice list: The New Keynesian AS–AD Model — qualifier level (Lista 7)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Parameters are symbols with the stated admissible ranges; answers are closed forms, conditions, signs and limits.

Notation (Benigno 2015). Lower-case letters are log deviations from the steady state; bars denote the long run. Households: $u(C) - v(L) + \beta\{u(\bar{C}) - v(\bar{L})\}$, $u = C^{1-\tilde{\sigma}^{-1}}/(1-\tilde{\sigma}^{-1})$, $v = L^{1+\eta}/(1+\eta)$, $\rho \equiv -\ln\beta$. Resources: $y = s_c c + g$, $\sigma \equiv \tilde{\sigma} s_c$. Firms: demand $Y(j) = (P(j)/P)^{-\theta}(C+G)$, technology $Y(j) = AL(j)$; a fraction α of prices is preset at P^e. Parameters: $0 < \alpha < 1$, $\sigma, \eta, \rho > 0$, $\theta > 1$; unless stated, $p^e = \bar{p} = 0$, $g = \bar{g} = 0$ and no taxes.

Block 1 — Foundations: AD, AS, natural and efficient output, taxes

(Benigno §§1–5 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) The slope κ of the AS curve $p - p^e = \kappa(y - y_n)$, in terms of the primitives, is:

A. $(1-\alpha)(\sigma^{-1}+\eta)$

B. $\alpha(\sigma^{-1}+\eta)/(1-\alpha)$

C. $(1-\alpha)(\sigma+\eta)$

D. $(1-\alpha)(\sigma^{-1}+\eta)/\alpha$

Answer: D

Step — derivation. Flexible firms set $\tilde{p} = w - a + \mu$, a markup over marginal cost W/A. Labor supply in logs, $w - p = \eta l + \tilde{\sigma}^{-1} c$, with $l = y - a$ (technology) and $c = (y-g)/s_c$ (resources), reads $w - p = \eta(y-a) + \sigma^{-1}(y-g)$, so $\tilde{p} - p = (\sigma^{-1}+\eta)(y - y_n)$. Aggregating, $p - p^e = (1-\alpha)[(\tilde{p}-p) + (p-p^e)]$, i.e. $\alpha(p-p^e) = (1-\alpha)(\sigma^{-1}+\eta)(y-y_n)$.

Step — distractors. A drops the feedback of p on the flexible firms' relative price, writing $p - p^e = (1-\alpha)(\tilde{p} - p)$; B swaps the weights of preset and flexible prices; C makes two mistakes: it drops the feedback, as A does, and reads σ as the curvature of utility, not the elasticity of substitution.

Ref: Benigno (2015), §4.1, eqs. (11)–(17), pp. 507–508.

b) (3 points) *AS–AD × taxation.* Short-run productivity rises ($a > 0$) and a payroll tax raises the (log) markup by $\mu > 0$, both temporarily. The changes in natural and efficient output, $(\Delta y_n,\ \Delta y_e)$, are:

A. $((a-\mu)/(\sigma^{-1}+\eta),\ a/(\sigma^{-1}+\eta))$

B. $([(1+\eta)a - \mu]/(\sigma^{-1}+\eta),\ (1+\eta)a/(\sigma^{-1}+\eta))$

C. $([(1+\eta)a - \mu]/(\sigma^{-1}+\eta),\ [(1+\eta)a - \mu]/(\sigma^{-1}+\eta))$

D. $(a/(\sigma^{-1}+\eta),\ a/(\sigma^{-1}+\eta))$

Answer: B

Step — derivation. Setting $\tilde{p} = p$ in the gap relation of (a): $(\sigma^{-1}+\eta)y_n = (1+\eta)a + \sigma^{-1}g - \mu$, and the payroll tax enters only through the aggregate markup (eq. 13). The planner sets $v_l(Y/A)/u_c(Y-G) = A$, with no markup and no tax: $\sigma^{-1}(y-g) + \eta(y-a) = a$. Productivity carries $1+\eta$ in both: it raises the marginal product (a) and saves hours ($l = y - a$).

Step — distractors. A uses $l = y$ and loses the ηa term; C lets the planner face the tax; D makes two mistakes: it uses $l = y$, as A does, and treats the rebated payroll tax as lump sum, so μ leaves y_n.

Ref: Benigno (2015), §4.1, eqs. (13), (15), pp. 507–508; §4.4, eq. (19), p. 509.

c) (2 points, T or F) Because θ sets the markup, a higher demand elasticity θ makes the AS curve steeper.

Answer: F

Step — proof. $\kappa = (1-\alpha)(\sigma^{-1}+\eta)/\alpha$ has no θ. It would enter only if a firm's marginal cost depended on its own output (decreasing returns, firm-specific labor: real rigidity); with $Y(j) = AL(j)$ and one labor market, marginal cost W/A is common to all firms. θ moves the level: $\mu_\theta = \ln \frac{\theta}{\theta-1}$ gives $\partial y_n/\partial\theta = \frac{1}{\theta(\theta-1)(\sigma^{-1}+\eta)} > 0$, a parallel shift of the AS. It reappears as the weight θ/κ on price stability in the loss function (33).

Ref: Benigno (2015), §4.1, eqs. (13), (17), pp. 507–508; §11, eq. (33), p. 521; Romer (2012), §§6.5 and 6.7.

d) (2 points, T or F) *AS–AD × fiscal news.* Holding i, $\bar{p}$ and p^e fixed, news that long-run government spending will be higher ($\bar{g}\uparrow$, g unchanged) lowers current output.

Answer: T

Step — proof. By (15), $\bar{y}_n - \bar{g} = \frac{(1+\eta)\bar{a}-\bar{\mu}-\eta\bar{g}}{\sigma^{-1}+\eta}$: long-run output absorbs only the share $\frac{\sigma^{-1}}{\sigma^{-1}+\eta} < 1$ of $\bar{g}$, so long-run consumption falls and smoothing cuts current demand; the AS does not move (y_n depends on current g only). Solving AD and AS, $\frac{dy}{d\bar{g}} = -\frac{\eta}{(\sigma^{-1}+\eta)(1+\sigma\kappa)} = -m_{\bar{g}} < 0$ for every $\eta > 0$; it vanishes only as $\eta \to 0$.

Ref: Benigno (2015), §8, Table 1, p. 515.

Question 2. (15 points) Discursive. *AS–AD foundations × taxation × supply shocks.* Build the model from its primitives (no taxes unless stated).

a) (3 points) From the household problem with budget $PC + \frac{\bar{P}\bar{C}}{1+i} = WL + \frac{\bar{W}\bar{L}}{1+i} + T$, derive the Euler equation and the labor-supply condition; log-linearize and combine with $y = s_c c + g$ to obtain the AD curve. Give its slope in the (y, p) plane, explain it, and sign the effect of each shifter.

Step — FOCs. $\mathcal{L} = u(C) - v(L) + \beta[u(\bar{C}) - v(\bar{L})] + \lambda\big[WL + \frac{\bar{W}\bar{L}}{1+i} + T - PC - \frac{\bar{P}\bar{C}}{1+i}\big]$, λ the marginal utility of nominal wealth: $u_c(C) = \lambda P$, $\beta u_c(\bar{C}) = \lambda\bar{P}/(1+i)$, $v_l(L) = \lambda W$. Dividing, $\frac{u_c(C)}{\beta u_c(\bar{C})} = (1+i)\frac{P}{\bar{P}} \equiv 1 + r$ (Euler) and $\frac{v_l(L)}{u_c(C)} = \frac{W}{P}$ (labor supply).

Step — log-linear Euler. With $u_c = C^{-1/\tilde{\sigma}}$, $\ln(1+i) \approx i$ and $\rho = -\ln\beta$: $\bar{c} - c = \tilde{\sigma}[i - (\bar{p} - p) - \rho]$.

Step — AD. Resources give $c = (y-g)/s_c$ and $\bar{c} = (\bar{y}-\bar{g})/s_c$; long-run prices are flexible, $\bar{y} = \bar{y}_n$; and $\sigma = \tilde{\sigma}s_c$:

$$y = \bar{y}_n + (g - \bar{g}) - \sigma[i - (\bar{p} - p) - \rho].$$

Slope $dp/dy = -1/\sigma$: given i and $\bar{p}$, a higher p lowers expected inflation $\bar{p} - p$, raising the real rate and saving (an intertemporal channel; no LM curve). Shifters: $\partial y/\partial i = -\sigma$, $\partial y/\partial g = 1$, $\partial y/\partial\bar{p} = \sigma$ and, through $\bar{y}_n$ in (15), $\partial y/\partial\bar{a} = \frac{1+\eta}{\sigma^{-1}+\eta}$, $\partial y/\partial\bar{\mu} = -\frac{1}{\sigma^{-1}+\eta}$, $\partial y/\partial\bar{g} = -\frac{\eta}{\sigma^{-1}+\eta}$.

Economic intuition. The AD is the Euler equation in output: higher expected income or a lower real rate raises demand today.

Ref: Benigno (2015), §3, eqs. (1)–(6), pp. 505–506; §5, eq. (21), p. 509; Kurlat (2020), §6.2 and §7.2.

Grading rubric.

- FOCs with the meaning of λ: 1 pt.
- Log-linear Euler: 1 pt. Quoted instead of derived: max. a third of the item.
- AD, slope and mechanism, signed shifters: 1 pt; slope via money demand: 0 for this line.

b) (3 points) From the firm's problem derive the optimal flexible price, natural output and the AS curve with a fraction α of preset prices.

Step — optimal price. $\Pi(j) = (P(j) - \frac{W}{A})(\frac{P(j)}{P})^{-\theta}(C+G)$ by demand (9) and technology; $\Pi'(P(j))$ has the sign of $(1-\theta)P(j) + \theta\frac{W}{A}$, positive below and negative above $P(j) = \frac{\theta}{\theta-1}\frac{W}{A}$, the unique maximum ($\theta > 1$). In logs $\tilde{p} = w - a + \mu$, $\mu = \ln\frac{\theta}{\theta-1}$.

Step — gap relation. Labor supply from (a) in logs, $w - p = \eta(y-a) + \sigma^{-1}(y-g)$ (using $l = y - a$, $\tilde{\sigma}^{-1}c = \sigma^{-1}(y-g)$), gives $\tilde{p} - p = (\sigma^{-1}+\eta)y - (1+\eta)a - \sigma^{-1}g + \mu = (\sigma^{-1}+\eta)(y - y_n)$, $y_n = \frac{(1+\eta)a + \sigma^{-1}g - \mu}{\sigma^{-1}+\eta}$: the output at which flexible firms keep their relative price.

Step — AS. $p = \alpha p^e + (1-\alpha)\tilde{p}$ gives $\alpha(p-p^e) = (1-\alpha)(\tilde{p}-p)$, so $p - p^e = \kappa(y - y_n)$, $\kappa = \frac{(1-\alpha)(\sigma^{-1}+\eta)}{\alpha}$, falling in α from ∞ ($\alpha \to 0$, vertical AS at y_n) to 0 ($\alpha \to 1$, flat AS at p^e).

Economic intuition. Real marginal cost rises with $y - y_n$ through the real wage (η, σ^{-1}); the $1-\alpha$ flexible firms pass it into prices.

Ref: Benigno (2015), §4.1, eqs. (9)–(17), pp. 506–508; Romer (2012), problem 6.16.

Grading rubric.

- Optimal price with the second-order argument: 1 pt.
- Gap relation and y_n: 1 pt.
- AS and κ: 1 pt. κ quoted: max. a third of the item.

c) (3 points) Derive efficient output from the planner's problem and compare it with natural output. Show how labor, payroll, sales and consumption taxes affect each, and find the sales subsidy that makes the steady state efficient. How does it depend on θ?

Step — planner. $\max_Y u(Y-G) - v(Y/A)$ is strictly concave; in its FOC $u_c(Y-G) = v_l(Y/A)/A$ the left side falls from ∞ to 0 and the right side rises without bound: one solution. In logs $\sigma^{-1}(y-g) + \eta(y-a) = a$, so $y_e = \frac{(1+\eta)a + \sigma^{-1}g}{\sigma^{-1}+\eta}$ and, by (b), $y_n - y_e = -\frac{\mu}{\sigma^{-1}+\eta} < 0$: the markup is a wedge between the marginal rate of substitution and the marginal product A.

Step — taxes. Profits (10) give $P(j) = \frac{\theta}{\theta-1}\frac{1+\tau_w}{1-\tau_y}\frac{W}{A}$; labor supply (7) is $\frac{v_l}{u_c} = \frac{1-\tau_l}{1+\tau_c}\frac{W}{P}$. Eliminating W/P with $P(j) = P$: $\frac{v_l(Y_n/A)}{u_c(Y_n-G)} = Ae^{-\mu}$, with gross markup $e^{\mu} = \frac{\theta}{\theta-1}\frac{(1+\tau_w)(1+\tau_c)}{(1-\tau_y)(1-\tau_l)}$ (eq. 13; μ stays the log markup, as in (b)), increasing in each tax: every tax lowers y_n (by $\frac{1}{\sigma^{-1}+\eta}$ per unit of μ); none enters the planner's condition, so y_e is untouched.

Step — efficient subsidy. $e^{\mu} = 1$ with a sales instrument only: $1 - \tau_y = \frac{\theta}{\theta-1}$, $\tau_y = -\frac{1}{\theta-1}$, financed lump sum (eq. 18); a payroll subsidy $\tau_w = -\frac{1}{\theta}$ also works. The rate $\frac{1}{\theta-1}$ falls in θ, $\to 0$ as $\theta \to \infty$ and $\to \infty$ as $\theta \to 1^+$.

Economic intuition. A markup taxes labor, holding the real wage below the marginal product; a subsidy that undoes it aligns y_n with y_e.

Ref: Benigno (2015), §4.1, eqs. (7), (10), (13); §4.4, eq. (19), p. 509; footnote 21, p. 522; Kurlat (2020), §9.2.

Grading rubric.

- Planner, uniqueness and y_e: 1 pt.
- Taxes in the markup, effect on y_n only: 1.5 pts. Taxes treated as lump sum: 0 for this line.
- Subsidy and its dependence on θ: 0.5 pt.

d) (3 points) Start from an efficient steady state ($y_n = y_e = 0$, as in (c)). An oil shock raises the markup by $\mu > 0$ and lowers productivity, $a < 0$, for the short run only. Find the output gap in terms of the natural real rate and the policy rate. Characterize y, $p - p^e$ and $y - y_e$ if the bank keeps $i = \rho$; find the i that delivers $p = p^e$ and its outcome. Draw the AS–AD diagram (old and new AS, AD, both equilibria, y_n and y_e after the shock).

Step — gap. AS (b) into AD (a), with $g = \bar{g}$: $(1+\sigma\kappa)(y - y_n) = \bar{y}_n - y_n - \sigma(i - \rho) + \sigma(\bar{p} - p^e)$. With the natural rate $r_n \equiv \rho + \sigma^{-1}(\bar{y}_n - y_n)$, the real rate at which demand equals natural output:

$$y - y_n = \frac{\sigma}{1+\sigma\kappa}\left[r_n - i + (\bar{p} - p^e)\right].$$

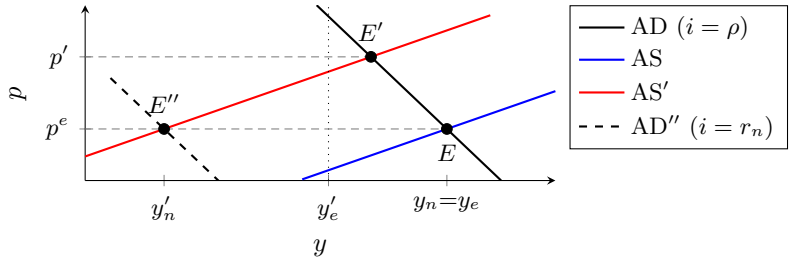
Step — the shock. By (b)–(c), with $\bar{y}_n = 0$: $y_n' = \frac{(1+\eta)a - \mu}{\sigma^{-1}+\eta} < y_e' = \frac{(1+\eta)a}{\sigma^{-1}+\eta} < 0$ and $r_n - \rho = -\sigma^{-1}y_n' = \frac{\mu - (1+\eta)a}{1+\sigma\eta} > 0$.

Step — $i = \rho$. The gap is $\frac{\bar{y}_n - y_n'}{1+\sigma\kappa} > 0$: $y = \frac{\sigma\kappa y_n' + \bar{y}_n}{1+\sigma\kappa} = \frac{\sigma\kappa}{1+\sigma\kappa}y_n' \in (y_n', 0)$, $p - p^e = -\frac{\kappa y_n'}{1+\sigma\kappa} > 0$ (stagflation), and

$$y - y_e' = \frac{(1+\eta)|a| - \sigma\kappa\mu}{(\sigma^{-1}+\eta)(1+\sigma\kappa)} \gtrless 0 \iff (1+\eta)|a| \gtrless \sigma\kappa\mu :$$

the productivity half lowers y'_e one for one with y'_n, while output absorbs only the share $\frac{\sigma\kappa}{1+\sigma\kappa}$ of the markup half (the figure draws the case $(1+\eta)|a| > \sigma\kappa\mu$).

Step — $p = p^e$. It requires $y = y'_n$, i.e. $i = r_n$ (with $\bar{p} = p^e$), a hike of $\frac{\mu-(1+\eta)a}{1+\sigma\eta}$; then $y - y'_e = -\frac{\mu}{\sigma^{-1}+\eta} < 0$. No i delivers both $p = p^e$ and $y = y'_e$: the markup half creates the trade-off.



Economic intuition. The productivity half moves y_n and y_e alike; the markup half moves only y_n. Both raise r_n, so $i = \rho$ sits below the price-stabilizing rate.

Ref: Benigno (2015), §§5–7, pp. 509–514; natural rate on p. 513.

Grading rubric.

- Gap formula derived: 1 pt. Shock reduced to one of its halves: max. half of the item.
- Both policies and the condition on the sign of $y - y'_e$: 1 pt; a sign without the condition: 0 for this line.
- Graph (scored on its own): axes, AS′ through (y'_n, p^e), AD and AD″, E', E'' and y'_e: 1 pt.

e) (3 points) *Parameter analysis.* For the markup shock alone ($\mu > 0$, $a = 0$, $i = \rho$), find y, $p - p^e$ and the natural gap $y - y_n$ in closed form, sign their responses to α, σ, η and μ with proofs in a table, give the limits $\alpha \to 0$ and $\alpha \to 1$, and read the results.

Step — closed forms. With $a = 0$: $y_e = 0$, $y_n = -\frac{\sigma\mu}{1+\sigma\eta}$. With $\xi \equiv \frac{1-\alpha}{\alpha}$ (free of σ, η), $\sigma\kappa = \xi(1+\sigma\eta)$ and $\Delta \equiv 1 + \xi(1+\sigma\eta)$, (d) at $i = \rho$ gives $y = \frac{\sigma\kappa y_n}{1+\sigma\kappa} = -\frac{\sigma\mu\xi}{\Delta}$, $p - p^e = -\frac{\kappa y_n}{1+\sigma\kappa} = \frac{\mu\xi}{\Delta}$, $y - y_n = \frac{\sigma\mu}{(1+\sigma\eta)\Delta}$.

Step — signs. Differentiating ($\partial_\xi y_n = 0$, $\partial_\sigma y_n = -\frac{\mu}{(1+\sigma\eta)^2}$, $\partial_\eta y_n = \frac{\sigma^2\mu}{(1+\sigma\eta)^2}$):

$$\begin{array}{lll} \partial_\xi y = -\frac{\sigma\mu}{\Delta^2}, & \partial_\xi(p-p^e) = \frac{\mu}{\Delta^2}, & \partial_\xi(y-y_n) = -\frac{\sigma\mu}{\Delta^2}, \\ \partial_\sigma y = -\frac{\mu\xi(1+\xi)}{\Delta^2}, & \partial_\sigma(p-p^e) = -\frac{\mu\eta\xi^2}{\Delta^2}, & \partial_\sigma(y-y_n) = \frac{\mu(1+\xi-\xi\sigma^2\eta^2)}{(1+\sigma\eta)^2\Delta^2}, \\ \partial_\eta y = \frac{\sigma^2\mu\xi^2}{\Delta^2}, & \partial_\eta(p-p^e) = -\frac{\sigma\mu\xi^2}{\Delta^2}, & \partial_\eta(y-y_n) = -\frac{\sigma^2\mu[1+2\xi(1+\sigma\eta)]}{(1+\sigma\eta)^2\Delta^2}. \end{array}$$

As $d\xi/d\alpha = -1/\alpha^2 < 0$, the α-signs are opposite to the ξ-signs (in κ: $\partial_\kappa \frac{\kappa}{1+\sigma\kappa} = \frac{1}{(1+\sigma\kappa)^2} > 0$); all results are proportional to μ; and $\partial_\sigma(y - y_n) > 0$ iff $1 + \xi > \xi\sigma^2\eta^2$, i.e. $\sigma^2\eta^2 < \frac{1}{1-\alpha}$.

	α	σ	η	μ
y_n	0	$-$	$+$	$-$
y $(= y - y_e)$	$+$	$-$	$+$	$-$
$p - p^e$	$-$	$-$	$-$	$+$
$y - y_n$	$+$	$+$ iff $\sigma\eta < (1-\alpha)^{-1/2}$	$-$	$+$

Step — limits. $\alpha \to 1$ ($\kappa \to 0$): $y \to 0$, $p \to p^e$, $y - y_n \to -y_n$ (output is demand-determined and the AD has not moved). $\alpha \to 0$ ($\kappa \to \infty$): $y \to y_n$, $p - p^e \to \frac{\mu}{1+\sigma\eta}$, $y - y_n \to 0$. Knife edge: $1 + \xi - \xi\sigma^2\eta^2$ falls in σ and vanishes once, at $\sigma^* = \frac{1}{\eta\sqrt{1-\alpha}}$, where the natural gap peaks.

Step — reading. With i held, demand pins output down: rigid prices shield output from a cost shock, flexible ones pass it into prices ($y \to y_n$). As y_e is fixed, more flexibility enlarges the efficiency loss and the price surprise while shrinking the natural gap, which is thus no welfare criterion. A

higher σ turns the price surprise into a larger fall in demand and deepens the fall of y_n (so the natural gap first widens, then narrows); a higher η shrinks the fall of y_n and steepens the AS, so output and prices move less.

Economic intuition. Output absorbs the share $\frac{\sigma\kappa}{1+\sigma\kappa}$ of the fall in y_n, rising in κ and σ.

Ref: Benigno (2015), §7, pp. 513–514; Romer (2012), problem 6.16.

Grading rubric.

- Closed forms: 1 pt.
- Signs proved, limits and the σ condition: 1 pt; a sign or limit without proof: 0 for that entry.
- Table and reading: 1 pt; signs without proofs or without reading: max. half of the item.

Block 2 — Policy: optimal rules, the zero bound and household debt

(Benigno §§6–12 • 25 points)

Question 3. (10 points) Objective items. Ranges: $\sigma, \eta, \kappa, \Delta > 0$; the long run is unchanged unless stated.

a) (3 points) *Monetary × fiscal policy.* A temporary cut Δ in the consumption tax (like the 2009 IPI cut on cars: $\tau_c = -\Delta$, $\bar{\tau}_c = 0$), entering the markup one for one, changes natural output y_n and the natural real rate r_n (the rate at which demand equals y_n) by, respectively:

A. $\left(\frac{\Delta}{\sigma^{-1}+\eta},\ \frac{\eta\Delta}{\sigma^{-1}+\eta}\right)$

B. $\left(\frac{\Delta}{\sigma^{-1}+\eta},\ -\frac{\sigma^{-1}\Delta}{\sigma^{-1}+\eta}\right)$

C. $\left(\frac{(1+\eta)\Delta}{\sigma^{-1}+\eta},\ \frac{(1-\sigma^{-1})\eta\,\Delta}{\sigma^{-1}+\eta}\right)$

D. $\left(\frac{\Delta}{\sigma^{-1}+\eta},\ \frac{(2\sigma^{-1}+\eta)\Delta}{\sigma^{-1}+\eta}\right)$

Answer: A

Step — supply channel. By (13) the markup falls by Δ, so by (15), $y_n = \frac{(1+\eta)a+\sigma^{-1}g-\mu}{\sigma^{-1}+\eta}$ rises by $\frac{\Delta}{\sigma^{-1}+\eta}$; $\bar{y}_n$ stays (the cut is temporary), so the term $\sigma^{-1}(\bar{y}_n - y_n)$ of r_n falls by $\frac{\sigma^{-1}\Delta}{\sigma^{-1}+\eta}$.

Step — demand channel. Taxed consumption enters the Euler equation, $\bar{c} - c = \tilde{\sigma}[i - (\bar{p} - p) - (\bar{\tau}_c - \tau_c) - \rho]$, and so the AD (21); setting $y = y_n$, $r_n = \rho + \sigma^{-1}(\bar{y}_n - y_n) + \sigma^{-1}(g - \bar{g}) + (\bar{\tau}_c - \tau_c)$. Consumption is cheaper today than tomorrow: the last term of r_n rises by Δ.

Step — net. $\Delta r_n = \Delta - \frac{\sigma^{-1}\Delta}{\sigma^{-1}+\eta} = \frac{\eta\Delta}{\sigma^{-1}+\eta} > 0$, zero only if $\eta = 0$: demand wins, so keeping $y = y_n$ needs a slightly higher i. B keeps the supply channel only (the tax treated as a payroll tax); C reads the markup cut as a productivity gain (coefficient $1 + \eta$ in (15)); D flips the sign of the supply channel (a higher current y_n read as raising r_n).

Ref: Benigno (2015), §4.1, eqs. (13) and (15); §5, eq. (21); §8.

b) (3 points) *Fiscal multipliers.* Let $\Omega \equiv (\sigma^{-1} + \eta)(1 + \sigma\kappa)$. With i, $\bar{p}$ and p^e fixed, a temporary rise in g ($\bar{g}$ unchanged, lump-sum financed) changes output and the gap $y - y_n$, per unit of g, by:

A. $\left(\frac{\sigma^{-1}+\eta}{\Omega},\ \frac{\sigma^{-1}+\eta}{\Omega}\right)$

B. $\left(\frac{\sigma^{-1}+\eta+\kappa}{\Omega},\ \frac{\sigma^{-1}+\eta+\kappa}{\Omega}\right)$

C. $\left(\frac{\sigma^{-1}+\kappa}{\Omega},\ 0\right)$

D. $\left(\frac{\sigma^{-1}+\eta+\kappa}{\Omega},\ \frac{\eta}{\Omega}\right)$

Answer: D

Step — both curves move. The AD (21) shifts right by dg and, by (15), the AS by $dy_n = \frac{\sigma^{-1}}{\sigma^{-1}+\eta}dg$. Substituting the AS into the AD, $(1 + \sigma\kappa)\,dy = dg + \sigma\kappa\,dy_n$, so $\frac{dy}{dg} = \frac{1}{1+\sigma\kappa} + \frac{\kappa}{\Omega} = \frac{\sigma^{-1}+\eta+\kappa}{\Omega}$.

Step — gap. $\frac{d(y-y_n)}{dg} = \frac{\sigma^{-1}+\eta+\kappa-\sigma^{-1}(1+\sigma\kappa)}{\Omega} = \frac{\eta}{\Omega} > 0$ (zero if $\eta = 0$: the curves shift in proportion), and $\frac{dy}{dg} - 1 = -\frac{\sigma\eta\kappa}{\Omega} < 0$. A forgets the AS shift (pure demand); B measures the gap from the old y_n; C uses flexible prices (output jumps to the new y_n).

Ref: Benigno (2015), §8, eqs. (22)–(23), Table 1.

c) (2 points, T or F) At the zero lower bound with $r_n < 0$, a credible commitment to a long-run price level higher by $-r_n$ restores $y = y_n$ and $p = p^e$, and the size of the commitment needed does not depend on the degree of price rigidity.

Answer: T

Step — proof. Substituting the AS into the AD, $y - y_n = \frac{\sigma}{1+\sigma\kappa}[r_n - i + (\bar{p} - p^e)]$; at $i = 0$ it vanishes iff $\bar{p} - p^e = -r_n$, for every κ, and then $p = p^e$ by the AS. And nothing in r_n (ρ, σ, fiscal terms, the flexible-price outputs $y_n, \bar{y}_n$) involves α: rigidity scales the slump, not the cure.

Ref: Benigno (2015), §9.

d) (2 points, T or F) *AS–AD × household debt.* In the borrower–saver economy of Question 4(d), with constant inflation between the short and the long run, more price flexibility reduces the output cost of a deleveraging shock.

Answer: F

Step — paradox of flexibility. With $\bar{p}_n - p$ fixed, only the Fisher term of (32) moves with p: $\frac{dy}{dp} = \frac{(1-\chi)d_0}{\chi} > 0$. With the AS and y_n unchanged, from a zero gap $y = \frac{(1-\chi)\hat{d}}{\chi-(1-\chi)d_0\kappa}$ and $\frac{\partial y}{\partial \kappa} = \frac{(1-\chi)^2 d_0\,\hat{d}}{[\chi-(1-\chi)d_0\kappa]^2} < 0$ for $\hat{d} < 0$: the fall grows with κ on the whole stable range (Question 4(e)).

Ref: Benigno (2015), §10, eq. (32).

Question 4. (15 points) Discursive. *Optimal policy × zero lower bound × household debt.* Ranges: $\sigma, \kappa, \rho > 0$, $\eta \geq 0$, $\theta > 1$; in (d)–(e), $\chi \in (0,1)$, $d_0 > 0$, $\beta \in (0,1)$. Shocks are temporary.

a) (3 points) The bank minimizes Benigno's welfare loss $L = \frac{1}{2}(y - y_e)^2 + \frac{\theta}{2\kappa}(p - p^e)^2$ (eq. 33) subject to the AS. Find the optimal relation between the efficient gap and price surprises, describe it in the (y, p) plane, and prove that shocks that leave μ unchanged create no trade-off.

Step — substitution. The bank picks y by moving i along the AD, and the AS ties $p - p^e = \kappa(y - y_n)$ to it: $L(y) = \frac{1}{2}(y - y_e)^2 + \frac{1}{2}\theta\kappa(y - y_n)^2$, with $L'' = 1 + \theta\kappa > 0$, so the FOC gives the unique minimum.

Step — targeting rule. FOC: $(y - y_e) + \theta\kappa(y - y_n) = 0$, i.e. $(y - y_e) + \theta(p - p^e) = 0$ (eq. 34), so $y^* = \frac{y_e + \theta\kappa y_n}{1+\theta\kappa}$, a weighted average of the targets. In the (y, p) plane the rule is the line through (y_e, p^e) with slope $-1/\theta$; the optimum is where it meets the AS (slope κ, through (y_n, p^e)).

Step — no trade-off. By (15) and (19), $y_n - y_e = -\frac{\mu}{\sigma^{-1}+\eta}$. If μ does not move ($\mu = 0$ around the efficient steady state), $y_n = y_e$ and (y_e, p^e) lies on both lines with $L = 0$, the global minimum (divine coincidence), reached with $i = r_n + (\bar{p} - p^e)$, which zeroes $y - y_n = \frac{\sigma}{1+\sigma\kappa}[r_n - i + (\bar{p} - p^e)]$.

Economic intuition. Welfare wants output at y_e, pricing ties prices to $y - y_n$; the bank trades the two gaps at the rate $\theta\kappa$, and only a markup shock pulls the targets apart.

Ref: Benigno (2015), §11, eqs. (33)–(34); §4.4, eq. (19).

Grading rubric.

- Substitution, FOC and second-order condition: 1.5 pts. Rule quoted: item max. 1 pt.
- Geometry of the rule: 0.5 pt.
- No trade-off proved from $y_n - y_e$: 1 pt; asserted without it: 0.

b) (3 points) A temporary cost-push shock raises the markup by $\mu > 0$ and moves productivity by a (either sign; Question 2(d) is the case $a < 0$); $\bar{y}_n = 0$, $\bar{p} = p^e$, $g = \bar{g}$. Find the optimal y, $p - p^e$ and i; compare them and the loss with strict price stability; say which part of the shock i offsets one for one. Show that no Taylor rule $i = \bar{\imath} + \psi_y(y - y_e) + \psi_\pi(p - p^e)$, $\psi_y, \psi_\pi \geq 0$, with a constant intercept $\bar{\imath}$ (a policy constant, not a long-run value) implements the optimum under all productivity shocks.

Step — optimum. With $M \equiv \frac{\mu}{\sigma^{-1}+\eta} = y_e - y_n$, (a) gives $y^* - y_n = \frac{M}{1+\theta\kappa} > 0$, $y^* - y_e = -\frac{\theta\kappa M}{1+\theta\kappa} < 0$, $p - p^e = \frac{\kappa M}{1+\theta\kappa} > 0$, between the pure policies ($-M$ and κM). Strict price stability ($y = y_n$, $p = p^e$) costs $L^{PS} = \frac{1}{2}M^2$; the optimum, $L^* = \frac{1}{2}\frac{\theta\kappa}{1+\theta\kappa}M^2$, so $L^*/L^{PS} = \frac{\theta\kappa}{1+\theta\kappa} < 1$.

Step — interest rate. By the gap formula with $\bar{p} = p^e$, $i^* = r_n - \frac{(\sigma^{-1}+\kappa)M}{1+\theta\kappa} < r_n$, the strict-price-stability rate, where $r_n = \rho + \frac{\sigma^{-1}[\mu-(1+\eta)a]}{\sigma^{-1}+\eta}$. The productivity part moves y_n and y_e equally and reaches i^* only through r_n: it is offset one for one. Against the passive $i = \rho$: $i^* - \rho = \frac{\kappa(\theta\sigma^{-1}-1)\mu - \sigma^{-1}(1+\eta)(1+\theta\kappa)a}{(\sigma^{-1}+\eta)(1+\theta\kappa)}$, positive if $a \leq 0$ and $\theta > \sigma$; with $a = 0$ the bank eases iff $\theta < \sigma$.

Step — Taylor rule. With $\mu = 0$, $y_n = y_e$, the rule is $i = \bar{\imath} + (\psi_y + \psi_\pi\kappa)(y - y_n)$ and the gap formula gives $y - y_n = \frac{r_n - \bar{\imath}}{\sigma^{-1}+\kappa+\psi_y+\psi_\pi\kappa}$: the optimum needs $\bar{\imath} = r_n$, yet $\frac{\partial r_n}{\partial a} = -\frac{\sigma^{-1}(1+\eta)}{\sigma^{-1}+\eta} \neq 0$: a fixed intercept fits one a at most, and feedback only shrinks the error (to zero as $\psi_\pi \to \infty$).

Economic intuition. A cost-push shock is shared between output and prices as $\theta\kappa : 1$; a productivity shock needs no sharing, only an i that follows r_n.

Ref: Benigno (2015), §11.

Grading rubric.

- Optimal y, p and the loss ratio: 1 pt.
- Optimal i, one-for-one offset, sign condition: 1 pt; a sign without the closed form: 0.
- Taylor-rule argument with the gap under the rule: 1 pt.

c) (3 points) Pessimism: long-run productivity is expected to fall, $\bar{a} < 0$, with $a = \mu = 0$ and $g = \bar{g}$. Find when the zero bound binds, the outcome at $i = 0$ with $\bar{p} = p^e$, and the $\bar{p}$ that restores the optimum; give the limits as $\kappa \to \infty$ and $\kappa \to 0$, and discuss credibility.

Step — natural rate. By (15) with bars, $\bar{y}_n = \frac{(1+\eta)\bar{a}}{\sigma^{-1}+\eta} < 0$, while $y_n = y_e = 0$: $r_n = \rho + \sigma^{-1}\bar{y}_n = \rho + \frac{(1+\eta)\bar{a}}{1+\sigma\eta}$. With $\mu = 0$ there is no trade-off and the optimum $y = 0$, $p = p^e$ needs $i = r_n$, which the bound forbids iff $r_n < 0$, i.e. $\bar{a} < -\frac{\rho(1+\sigma\eta)}{1+\eta}$: pessimism must outweigh impatience.

Step — at $i = 0$. The gap $\frac{\sigma(r_n - i)}{1+\sigma\kappa}$ is negative and closest to zero at $i = 0$: $y = \frac{\sigma r_n}{1+\sigma\kappa} < 0$, $p - p^e = \frac{\sigma\kappa r_n}{1+\sigma\kappa} < 0$, a slump with deflation. The bound caps the nominal rate, not the real one: $\bar{p} - p^e = -r_n > 0$ zeroes $r_n + (\bar{p} - p^e)$, so $y = 0$, $p = p^e$ at $i = 0$.

Step — limits and credibility. As $\kappa \to \infty$, $\frac{\sigma\kappa}{1+\sigma\kappa} \to 1$: $y \to 0$, $p - p^e \to r_n$ (prices absorb the shock); as $\kappa \to 0$, $p \to p^e$ and $y \to \sigma r_n$ (output does). Once the slump is over, a price-stability bank would rather set $\bar{p} = p^e$: the promise is time-inconsistent and works only if believed.

Economic intuition. Pessimism makes households want to save; only a negative real rate clears that wish, and at $i = 0$ it must come from expected inflation, which the bank can only promise.

Ref: Benigno (2015), §6.3 and §9.

Grading rubric.

- r_n and the binding condition: 1 pt.
- Outcome at $i = 0$ and the required $\bar{p}$: 1 pt.
- Limits from the closed forms, and credibility: 1 pt; limits without the closed form: 0.

d) (3 points) With borrowers and savers (share χ of savers, initial debt d_0), Benigno's AD is

$$y = \bar{y}_n - \varphi[i - (\bar{p}_n - p) - \rho] + \tfrac{1}{\chi}(g - \bar{g}) + \tfrac{1-\chi}{\chi}[\hat{d} + d_0(p - p^e)], \qquad \varphi \equiv \frac{\sigma\chi + (1-\chi)d_0\beta}{\chi},$$

where $\bar{p}_n$ is the long-run price level ($\bar{p}$ above) and $\hat{d}$ the change in the debt limit. Find dy/dp when $\bar{p}_n$ is anchored and when inflation $\bar{p}_n - p$ is constant, naming the two channels. Assuming the anchored AD slopes down and, under constant inflation, the AD is steeper than the AS, derive the spending multipliers m_A, m_C (i fixed), order them against $1/\chi$, and find when $m_A > 1$.

Step — slopes. Anchored: $\frac{dy}{dp} = -\varphi + \frac{(1-\chi)d_0}{\chi} = -\varpi$, $\varpi \equiv \sigma - \frac{d_0(1-\beta)(1-\chi)}{\chi}$. Real-rate channel: a higher p lowers expected inflation and raises the real rate, cutting savers' consumption ($-\sigma$) and what

borrowers may borrow ($-\frac{(1-\chi)d_0\beta}{\chi}$). Fisher channel: it erodes inherited nominal debt ($+\frac{(1-\chi)d_0}{\chi}$). As $\beta<1$, $\varpi<\sigma$: debt steepens the AD, which slopes down iff $\varpi>0$. Constant inflation: the real rate no longer moves with p, and $\frac{dy}{dp}=\frac{(1-\chi)d_0}{\chi}>0$.

Step — multipliers. A temporary g shifts the AD by $\frac{dg}{\chi}$ and, by (15), raises y_n by $s\,dg$, $s\equiv\frac{1}{1+\sigma\eta}\le 1$; the AS gives $dp=\kappa(dy-s\,dg)$. Anchored: $dy=\frac{dg}{\chi}-\varpi\kappa(dy-s\,dg)$, so $m_A=\frac{1/\chi+\varpi\kappa s}{1+\varpi\kappa}$. Constant inflation, with $x\equiv(1-\chi)d_0\kappa$: $\chi\,dy=dg+x(dy-s\,dg)$, so $m_C=\frac{1-xs}{\chi-x}$, for $x<\chi$ (AD steeper than AS).

Step — ordering. m_A averages $\frac{1}{\chi}$ and $s<\frac{1}{\chi}$ with weight $\frac{\varpi\kappa}{1+\varpi\kappa}\in(0,1)$ on s, so $s<m_A<\frac{1}{\chi}$; and $m_C-\frac{1}{\chi}=\frac{x(1-\chi s)}{\chi(\chi-x)}>0$: $m_C>\frac{1}{\chi}>m_A$. Next, $m_A>1$ iff $\varpi\kappa(1-s)<\frac{1-\chi}{\chi}$, i.e. $\varpi\kappa<\frac{(1-\chi)(1+\sigma\eta)}{\chi\sigma\eta}$ (always if $\eta=0$), while $m_C-1=\frac{(1-\chi)+x(1-s)}{\chi-x}>0$. As $\chi\to1$, $\varpi\to\sigma$ and $m_A\to\frac{1+\sigma\kappa s}{1+\sigma\kappa}=\frac{\sigma^{-1}+\eta+\kappa}{\Omega}$, the multiplier of Question 3(b).

Economic intuition. Borrowers spend current income: the AD holds a Keynesian cross with multiplier $1/\chi$. Spending raises prices; with $\bar p_n$ anchored that raises the real rate and crowds out ($m_A<1/\chi$), with constant inflation it only deflates real debt and adds demand ($m_C>1/\chi$).

Ref: Benigno (2015), §10, eq. (32) and footnote 19.

Grading rubric.

- Both slopes with the two channels: 1 pt.
- Both multipliers derived: 1 pt (quoted: item max. 1 pt; §8 multiplier used: item max. 1.5 pts).
- Ordering and the condition for $m_A>1$, proved: 1 pt.

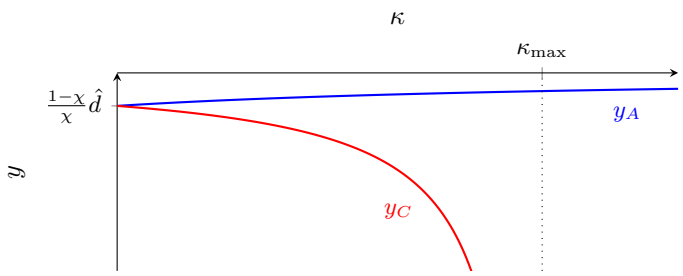
e) (3 points) *Parameter analysis.* A deleveraging shock lowers the debt limit, $\hat d<0$, with i, g fixed and the economy initially at $y=y_n=0$, $p=p^e$. Find each regime's output response and the stability bound; sign both responses and both multipliers of (d) against κ, d_0, χ, σ (given κ) in a table, with proofs; give the limits in κ and the knife-edges; draw both responses against κ and read them.

Step — closed forms and bound. $\hat d$ leaves y_n unchanged (natural output does not depend on who holds the debt), so the AS is $p-p^e=\kappa y$. Anchored: $y(1+\varpi\kappa)=\frac{1-\chi}{\chi}\hat d$, so $y_A=\frac{(1-\chi)\hat d}{\chi(1+\varpi\kappa)}$. Constant inflation: $\chi y=(1-\chi)(\hat d+d_0\kappa y)$, so $y_C=\frac{(1-\chi)\hat d}{\chi-x}$, negative only for $\kappa<\kappa_{\max}\equiv\frac{\chi}{(1-\chi)d_0}$ and unbounded as $\kappa\uparrow\kappa_{\max}$: the stability bound, where the AD slope in the (y,p) plane, $\frac{\chi}{(1-\chi)d_0}$, equals the AS slope κ (footnote 19).

Step — signs. Let $\ell_j\equiv-y_j>0$. (i) ℓ_A falls with $\varpi\kappa$; ϖ rises with σ and falls with d_0; and $\chi(1+\varpi\kappa)=\chi(1+\sigma\kappa)-(1-\beta)d_0(1-\chi)\kappa$ rises with χ while $1-\chi$ falls. (ii) $\frac{\partial\ell_C}{\partial\kappa}=\frac{(1-\chi)^2d_0|\hat d|}{(\chi-x)^2}>0$, $\frac{\partial\ell_C}{\partial d_0}=\frac{(1-\chi)^2\kappa|\hat d|}{(\chi-x)^2}>0$, $\frac{\partial\ell_C}{\partial\chi}=-\frac{|\hat d|}{(\chi-x)^2}<0$; σ is absent. (iii) $\frac{\partial m_A}{\partial(\varpi\kappa)}=\frac{s-1/\chi}{(1+\varpi\kappa)^2}<0$, $\frac{\partial m_A}{\partial s}=\frac{\varpi\kappa}{1+\varpi\kappa}>0$, $\frac{\partial m_A}{\partial(1/\chi)}>0$, and s falls with σ (if $\eta>0$). (iv) By (d), m_C rises with x and, as $\frac{\partial m_C}{\partial s}=-\frac{x}{\chi-x}<0$, falls with s; $\frac{\partial m_C}{\partial\chi}=-\frac{1+d_0\kappa(1-s)}{(\chi-x)^2}<0$.

	κ	d_0	χ	σ
ℓ_A	$-$	$+$	$-$	$-$
ℓ_C	$+$	$+$	$-$	0
m_A	$-$	$+$	$-$	$-$
m_C	$+$	$+$	$-$	$+$

σ at given κ; $\partial m_C/\partial\sigma=0$ if $\eta=0$.



Step — limits. $\kappa\to0$: $y_A,y_C\to\frac{1-\chi}{\chi}\hat d$ and $m_A,m_C\to\frac{1}{\chi}$ (prices do not move). $\kappa\to\infty$: $y_A\to0$, $m_A\to s$. $\kappa\uparrow\kappa_{\max}$: $\ell_C,m_C\to\infty$. Knife-edges: $\chi\to1$ removes the debt channel ($y_A,y_C\to0$, $m_C\to1$); $\varpi=0$, i.e. $d_0=\frac{\sigma\chi}{(1-\beta)(1-\chi)}$, makes the anchored AD vertical, so y_A and $m_A=\frac{1}{\chi}$ do not depend on κ.

Economic intuition. With $\bar{p}_n$ anchored, a lower p raises expected inflation and cuts the real rate: flexibility cushions the slump. With constant inflation it lowers future prices too and raises real debt: flexibility deepens the slump (paradox of flexibility). ℓ_C and m_C share the denominator $\chi - x$: spending is most powerful where the slump is deepest.

Ref: Benigno (2015), §10, eq. (32) and footnotes 19–20.

Grading rubric.

- Closed forms and the stability bound: 1 pt.
- Signs proved, limits, reading: 1 pt (signs unproved or unread: item max. 1.5 pts).
- Graph, scored on its own: labels, common intercept, y_A rising to 0, y_C diverging at $\kappa_{\max}$: 1 pt.

Block 3 — Integrative: a pandemic in the AS–AD model

(Benigno §§4–11; Kurlat chs. 6–7 • 25 points)

Question 5. (10 points) Objective items.

a) (3 points) *Expectations × AS–AD.* To fight a pandemic slump with i fixed, the central bank credibly raises the long-run price level by $d\bar{p} > 0$; the preset prices p^e are predetermined. Current output changes by:

A. $dy = \sigma\, d\bar{p}$

B. $dy = \frac{\sigma}{1+\sigma\kappa}\, d\bar{p}$

C. $dy = d\bar{p}$

D. $dy = \frac{\sigma\kappa}{1+\sigma\kappa}\, d\bar{p}$

Answer: B

Step — solve. AD (21) with $g = \bar{g}$ and i fixed: $dy = \sigma(d\bar{p} - dp)$, because expected inflation $\bar{p} - p$ lowers the real rate. AS (20) with p^e and y_n fixed: $dp = \kappa\, dy$. Hence $dy = \frac{\sigma}{1+\sigma\kappa} d\bar{p} > 0$ and $dp = \frac{\sigma\kappa}{1+\sigma\kappa} d\bar{p} \in (0, d\bar{p})$.

Step — distractors. A drops the AS feedback: it is the rigid-price limit $\kappa \to 0$. C makes two mistakes: it drops σ and the feedback, moving output one for one with $\bar{p}$. D is dp, the price response. As $\kappa \to \infty$, $dy \to 0$ and $dp \to d\bar{p}$: the promise then moves only prices.

Ref: Benigno (2015), §5, eqs. (20)–(21) and p. 510 (a higher $\bar{p}$ shifts the AD right); §9.

b) (3 points) *Growth × AS–AD.* Productivity rises permanently, $a = \bar{a} > 0$ (say, lasting gains from remote work), while the bank keeps $i = \rho$ and $\bar{p} = p^e$. The new equilibrium $(y,\ p - p^e)$ is:

A. $(0,\ -\frac{\kappa(1+\eta)a}{\sigma^{-1}+\eta})$

B. $(\frac{(1+\eta)a}{\sigma^{-1}+\eta},\ -\frac{\kappa(1+\eta)a}{\sigma^{-1}+\eta})$

C. $(\frac{(1+\eta)a}{\sigma^{-1}+\eta},\ 0)$

D. $(a,\ 0)$

Answer: C

Step — natural levels. The flexible price $\tilde{p} = w - a + \mu$ and labor supply $w - p = \eta(y-a) + \sigma^{-1}(y-g)$ give, at $g = \mu = 0$, $\tilde{p} - p = (\sigma^{-1} + \eta)y - (1+\eta)a$, so $y_n = \frac{(1+\eta)a}{\sigma^{-1}+\eta}$. With $\bar{a} = a$, $\bar{y}_n = y_n$ and $r_n = \rho + \sigma^{-1}(\bar{y}_n - y_n) = \rho$.

Step — equilibrium. The AS shifts right through (y_n, p^e) and the AD shifts right by $\bar{y}_n$, through the same point. AS (20) in AD (21) gives the gap $y - y_n = \frac{\sigma}{1+\sigma\kappa}[r_n - i + (\bar{p} - p^e)] = 0$, so $y = y_n$ and $p = p^e$.

Step — distractors. A holds output at its old level, as if the unchanged i fixed demand, and reads p off the new AS. B takes the AS's downward shift, κy_n, for the change in p. D is the knife-edge

$\sigma = 1$ (log utility with $s_c = 1$): $y_n - a = \frac{(1-\sigma^{-1})a}{\sigma^{-1}+\eta}$ vanishes only there, where income and substitution effects on hours cancel.

Ref: Benigno (2015), §4, eq. (15); §6.2 and p. 513 (a permanent shock leaves the natural real rate unchanged).

c) (2 points, T or F) *Consumption theory × AS–AD.* Hold i and the long-run price level fixed. A lump-sum transfer financed by public debt (such as Brazil's 2020 *auxílio emergencial*) leaves output unchanged in Benigno's representative-agent model, whereas in his borrower–saver model (eq. (32), stable equilibrium) a debt-financed transfer to the borrowers raises output.

Answer: T

Step — representative agent. The budget (2) holds only the present value T of net transfers. A transfer B now, repaid by taxes $(1+i)B$ later, changes it by $B - \frac{(1+i)B}{1+i} = 0$, so no choice changes at given prices. Lump-sum taxes enter neither the AD (21) nor the markup in the AS: y and p do not move (Ricardian equivalence).

Step — borrowers and savers. Benigno's AD is

$$y = \bar{y}_n - \varphi[i - (\bar{p}_n - p) - \rho] + \tfrac{1}{\chi}(g - \bar{g}) - \tfrac{1-\chi}{\chi}(\tau_b - \bar{\tau}_b) + \tfrac{1-\chi}{\chi}[\hat{d} + d_0(p - p^e)],$$

with $\varphi \equiv \sigma + (1-\chi)d_0\beta/\chi$. Borrowers stay at their debt limit and consume their current resources, so taxes enter only through $\tau_b - \bar{\tau}_b$; savers' taxes do not enter. A transfer now ($d\tau_b < 0$), repaid later by borrowers ($d\bar{\tau}_b > 0$) or by savers ($d\bar{\tau}_b = 0$), shifts the AD right by $\frac{1-\chi}{\chi}\Delta$, $\Delta \equiv -(d\tau_b - d\bar{\tau}_b) > 0$. With i and $\bar{p}_n$ fixed, $dy = \frac{1-\chi}{\chi}\Delta - \varpi\, dp$, where $\varpi \equiv \varphi - \frac{(1-\chi)d_0}{\chi} = \sigma - \frac{d_0(1-\beta)(1-\chi)}{\chi}$ nets the real-rate channel against the Fisher channel; the AS gives $dp = \kappa\, dy$ (y_n does not depend on the distribution of wealth). So $dy = \frac{(1-\chi)\Delta}{\chi(1+\varpi\kappa)} > 0$: stability means $1 + \varpi\kappa > 0$, which is automatic when $\varpi \geq 0$.

Ref: Benigno (2015), §3, eq. (2) and footnote 3; §10, eq. (32), pp. 519–521; Kurlat (2020), ch. 6, §6.2 (Taxes and Ricardian Equivalence) and Exercise 6.5.

d) (2 points, T or F) *Labor × AS–AD.* Okun's law maps gaps into unemployment, $u - u_n = -\gamma(y - y_n)$ with $\gamma > 0$. After the oil shock of Question 2(d) (short run only: the markup rises, $\mu > 0$, and productivity falls, $a < 0$), a bank that keeps $i = \rho$ and $\bar{p} = p^e$ leaves unemployment below its natural rate even though output falls.

Answer: T

Step — natural levels. The long run is untouched, $\bar{y}_n = 0$, while $y_n = \frac{(1+\eta)a - \mu}{\sigma^{-1}+\eta} < 0$. The natural real rate rises, $r_n - \rho = \sigma^{-1}(\bar{y}_n - y_n) > 0$, so holding $i = \rho$ is a loose policy.

Step — gap and output. AS (20) in AD (21): $y - y_n = \frac{\sigma}{1+\sigma\kappa}(r_n - \rho) = \frac{\bar{y}_n - y_n}{1+\sigma\kappa} > 0$, hence $u - u_n = -\gamma\frac{\bar{y}_n - y_n}{1+\sigma\kappa} < 0$; and $y = \frac{\sigma\kappa y_n + \bar{y}_n}{1+\sigma\kappa} = \frac{\sigma\kappa}{1+\sigma\kappa}y_n \in (y_n, 0)$.

Step — conditions. In general $u < u_n$ iff $\bar{y}_n > y_n$ (that is, $r_n > i$), and $y < 0$ iff $\sigma\kappa y_n + \bar{y}_n < 0$. The temporary shock, $y_n < 0 = \bar{y}_n$, meets both; a permanent one ($\bar{y}_n = y_n$) would leave $u = u_n$.

Economic intuition. Output falls with its natural level, but by less: the expected recovery raises the natural real rate, and a bank that does not follow it lets demand hold output above y_n.

Ref: Benigno (2015), §7 and p. 513 (natural real rate); Jones (2020), ch. 9, §9.4 (Okun's law).

Question 6. (15 points) Discursive. *Integrative: supply shocks × household debt × fiscal transfers × optimal policy.* A stylized pandemic hits the borrower–saver economy of Benigno §10: a share $\chi \in (0,1)$ of savers with discount factor $\beta \in (0,1)$, $\rho = -\ln\beta$; borrowers at their debt limit, with initial debt $d_0 > 0$; utility $u(C_j) = 1 - e^{-zC_j}$, $z > 0$, with $\sigma \equiv 1/(z\tilde{Y})$ ($\tilde{Y}$: steady-state output), and $v(L_j) = L_j^{1+\eta}/(1+\eta)$; Cobb–Douglas indices $L = L_s^{\chi}L_b^{1-\chi}$ and $W = W_s^{\chi}W_b^{1-\chi}$. For the short run only, productivity falls ($a < 0$) and the markup rises ($\mu > 0$); households expect lower long-run productivity ($\bar{a} < 0$); banks tighten the debt limit ($\hat{d} < 0$). Long-run prices stay anchored, $\bar{p}_n = p^e = 0$; initially $g = \bar{g} = 0$ and $\tau_b = \bar{\tau}_b = 0$, where τ_b is the lump-sum tax on each borrower (debt and taxes in shares of $\tilde{Y}$). Benigno's AD (32) reads

$$y = \bar{y}_n - \varphi\big[i - (\bar{p}_n - p) - \rho\big] + \tfrac{1}{\chi}\big[(g - \bar{g}) - (1-\chi)(\tau_b - \bar{\tau}_b)\big] + \tfrac{1-\chi}{\chi}\big[\hat{d} + d_0(p - p^e)\big],$$

with $\varphi \equiv [\sigma\chi + (1-\chi)d_0\beta]/\chi$.

a) (3 points) *Supply side.* Show that the AS curve $p - p^e = \kappa(y - y_n)$, natural output and efficient output do not depend on how consumption is split between the two types. Find y_n, y_e and $\bar{y}_n$ after the shocks, and say which shock moves which of them and which curve it shifts.

Step — aggregation — each type's labor condition $v_l(L_j)/u_c(C_j) = W_j/P$, the two indices, goods clearing. Type j sets $L_j^{\eta} e^{zC_j}/z = W_j/P$. Raising the savers' condition to the power χ, the borrowers' to $1-\chi$, and multiplying: $L^{\eta} e^{z[\chi C_s + (1-\chi)C_b]}/z = W/P$, and goods clearing gives $\chi C_s + (1-\chi)C_b = Y - G$. Only aggregate consumption enters, so a lump-sum reallocation between the types leaves aggregate labor supply unchanged. In log deviations, with $\sigma^{-1} = z\tilde{Y}$: $w - p = \eta l + \sigma^{-1}(y - g)$, the representative-agent condition.

Step — natural output and AS — flexible price $\tilde{p} = w - a + \mu$, $l = y - a$, $g = 0$. $\tilde{p} - p = \eta(y-a) + \sigma^{-1}y - a + \mu = (\sigma^{-1} + \eta)(y - y_n)$, with

$$y_n = \frac{(1+\eta)a - \mu}{\sigma^{-1} + \eta};$$

with a share α of preset prices, $p = \alpha p^e + (1-\alpha)\tilde{p}$ gives $p - p^e = \kappa(y - y_n)$, $\kappa = (1-\alpha)(\sigma^{-1} + \eta)/\alpha$. Neither χ nor d_0 appears.

Step — efficient and long-run output. Without the markup wedge ($\mu = 0$: aggregate MRS = MPL), $y_e = \frac{(1+\eta)a}{\sigma^{-1}+\eta}$, so $y_e - y_n = \frac{\mu}{\sigma^{-1}+\eta} > 0$. In the long run $\bar{\mu} = \bar{g} = 0$: $\bar{y}_n = \frac{(1+\eta)\bar{a}}{\sigma^{-1}+\eta} < 0$.

Step — which shock moves what. $a < 0$ lowers y_n and y_e by the same $\frac{(1+\eta)|a|}{\sigma^{-1}+\eta}$; $\mu > 0$ lowers y_n only and opens the wedge. Both shift the AS up and to the left (it crosses p^e at y_n). $\bar{a} < 0$ lowers $\bar{y}_n$ only, which sits in the AD; $\hat{d} < 0$ moves none of the three and shifts only the AD. The pandemic is a supply shock with a trade-off ($y_e > y_n$) plus two demand shocks.

Economic intuition. Exponential utility aggregates marginal utilities into total consumption, and the Cobb–Douglas index ties each type's labor income to total income: who consumes does not change how much is supplied, so distribution acts only on demand. That is what gives transfers their role in (c).

Ref: Benigno (2015), §4, eqs. (15), (17), (19); §10, eqs. (24)–(29).

Grading rubric.

- Aggregation by geometric averaging: 1 pt. Distribution neutrality asserted without it: 0 for this line.
- y_n, κ, y_e, $\bar{y}_n$ derived: 1 pt. Quoted: max. a third.
- Which shock moves which variable and which curve: 1 pt.

b) (3 points) *Demand and equilibrium.* Find the slope of the AD curve, the condition under which it slopes down and the channels behind it; the equilibrium y and p; the natural and efficient gaps as functions of i; the rate i_n that closes the natural gap; and the condition under which the zero lower bound $i \geq 0$ rules i_n out.

Step — AD — eq. (32) with $\bar{p}_n = p^e = 0$, $g = \bar{g}$, $\tau_b = \bar{\tau}_b$. $y = \zeta - \varpi p$, where $\zeta \equiv \bar{y}_n - \varphi(i-\rho) + \frac{1-\chi}{\chi}\hat{d}$ is demand at $p = p^e$ and $\varpi \equiv \varphi - \frac{(1-\chi)d_0}{\chi} = \sigma - \frac{d_0(1-\beta)(1-\chi)}{\chi}$. A higher p raises the real rate, so savers consume less and borrowers can carry less debt ($-\varphi$); it also erodes inherited nominal debt (Fisher channel, $+\frac{(1-\chi)d_0}{\chi}$). As $\chi > 0$, the AD slopes down iff $\varpi > 0 \iff \sigma\chi > d_0(1-\beta)(1-\chi)$. Maintained below: $1 + \varpi\kappa > 0$, implied by $\varpi > 0$ (stability, part (e)).

Step — equilibrium — AD and AS. With $p - p^e = \kappa(y - y_n)$: $(1 + \varpi\kappa)y = \zeta + \varpi\kappa y_n$. The system is linear with determinant $1 + \varpi\kappa \neq 0$, so

$$y = \frac{\zeta + \varpi\kappa y_n}{1 + \varpi\kappa}, \qquad p - p^e = \frac{\kappa(\zeta - y_n)}{1 + \varpi\kappa}$$

is the unique equilibrium.

Step — gaps — define i_n by $\zeta(i_n) = y_n$. Then $\zeta - y_n = \varphi(i_n - i)$, and

$$i_n = \rho + \frac{1}{\varphi}\Big[\bar{y}_n - y_n + \frac{1-\chi}{\chi}\hat{d}\Big], \qquad y - y_n = \frac{\varphi(i_n - i)}{1 + \varpi\kappa}, \qquad y - y_e = \frac{\varphi(i_n - i)}{1 + \varpi\kappa} - \frac{\mu}{\sigma^{-1} + \eta}.$$

Step — zero bound — part (a). $\varphi(i_n - \rho) = \frac{(1+\eta)(\bar{a}-a)+\mu}{\sigma^{-1}+\eta} + \frac{1-\chi}{\chi}\hat{d}$: the supply shocks ($a < 0$, $\mu > 0$) raise i_n; pessimism and deleveraging lower it, so its sign is ambiguous. As $\varphi > 0$, the bound rules i_n out iff $i_n < 0$, i.e. iff $\frac{(1+\eta)(\bar{a}-a)+\mu}{\sigma^{-1}+\eta} + \frac{1-\chi}{\chi}\hat{d} < -\varphi\rho$: the demand shocks dominate. Then at $i = 0$ both gaps are negative and $p - p^e = \frac{\kappa\varphi i_n}{1+\varpi\kappa} < 0$: prices fall in spite of the cost-push shock.

Economic intuition. The natural rate records a race: the supply shocks call for a higher rate, pessimism and deleveraging for a lower one. Only when demand loses by more than $\varphi\rho$ does the bound bind, and then prices fall in the middle of a cost shock.

Ref: Benigno (2015), §9; §10, eq. (32) and footnote 19; Romer (2012), §11.6.

Grading rubric.

- Slope, condition and both channels: 1 pt. Slope $-\sigma$ of the one-type model used: max. half of (b).
- Equilibrium, uniqueness and both gaps derived: 1 pt.
- i_n, its sign decomposition and the zero-bound condition: 1 pt. "The bound binds" asserted without the condition: 0 for this line.

c) (3 points) *Fiscal transfers.* The government pays Δ to each borrower ($\tau_b = -\Delta$, $\bar{\tau}_b$ unchanged) and taxes the savers in the long run to repay the debt. From the borrowers' budget constraint, derive the output effect per unit of aggregate transfer; find the transfer Δ_e that closes the efficient gap at $i = \rho$; find the effect of a transfer to savers financed the same way; and study $\chi \to 1$.

Step — Keynesian cross — borrower budget (25) at the limit (27), $W_b L_b + \Pi_b = PY$, savers' Euler equation (30). A borrower at the limit consumes $C_b = \frac{D}{1+r} - \frac{(1+i_0)B_{0,b}}{P} + Y - T_b$, with D the cap on real debt service $(1+r)B_b/P$ in (27) ($d_0 = D_0/\tilde{Y}$, $\hat{d} = (D - D_0)/\tilde{Y}$), $B_{0,b}$ inherited nominal debt and i_0 its rate: extra income and transfers are spent one for one, while savers' consumption depends only on the real rate. In (31), $y = \bar{y}_n + (g - \bar{g}) - \chi\sigma(r - \rho) + (1-\chi)\frac{C_b - \bar{C}_b}{\tilde{Y}}$, so at given i and p: $dy = (1-\chi)(dy + \Delta)$, i.e. $dy = \frac{1-\chi}{\chi}\Delta$. The propensity to spend out of aggregate income is $1 - \chi$; per unit of aggregate transfer $(1-\chi)\Delta$ the fixed-price multiplier is $\frac{1}{\chi}$.

Step — with prices — AD shift $\frac{1-\chi}{\chi}\Delta$; AS unmoved by (a). $dy = \frac{1-\chi}{\chi}\Delta - \varpi\, dp$ and $dp = \kappa\, dy$ give

$$m_T \equiv \frac{dy}{(1-\chi)\Delta} = \frac{1}{\chi(1 + \varpi\kappa)}, \qquad \frac{dp}{(1-\chi)\Delta} = \kappa m_T.$$

Per unit, a purchase g shifts the AD as much ($\frac{1}{\chi}$) but also raises y_n by $\frac{1}{1+\sigma\eta}$; its multiplier exceeds m_T by $\frac{\varpi\kappa}{(1+\sigma\eta)(1+\varpi\kappa)}$, positive iff $\varpi > 0$.

Step — efficient gap at $i = \rho$ — part (b) with ζ raised by $\frac{1-\chi}{\chi}\Delta$. $y = y_e$ iff $\zeta - y_n = (1 + \varpi\kappa)(y_e - y_n)$, which gives

$$\Delta_e = -\hat{d} + \frac{\chi}{1-\chi} \cdot \frac{(1+\eta)(a - \bar{a}) + \varpi\kappa\mu}{\sigma^{-1} + \eta}.$$

It replaces the lost credit one for one ($-\hat{d}$), offsets pessimism net of the fall in y_e ($a - \bar{a}$), and pushes y past y_n to close the markup wedge, at the cost $p - p^e = \frac{\kappa\mu}{\sigma^{-1}+\eta} > 0$. Δ_e has the sign of $(1-\chi)|\hat{d}|(\sigma^{-1} + \eta) + \chi[(1+\eta)(a - \bar{a}) + \varpi\kappa\mu]$, positive whenever $a \geq \bar{a}$ and $\varpi > 0$.

Step — transfer to savers. It changes T_s and $\bar{T}_s$ only. Savers' consumption obeys (30), which contains only r, and the borrowers' budgets do not contain T_s; so (31), (32), ζ, y and p do not move. A saver receives a transfer and pays a tax of equal present value: Ricardian equivalence holds for the unconstrained type.

Step — $\chi \to 1$. $\varpi \to \sigma$, so $m_T \to \frac{1}{1+\sigma\kappa}$, finite. But raising y by dy takes $\Delta = \frac{\chi(1+\varpi\kappa)}{1-\chi}dy \to \infty$ per borrower, and at $\chi = 1$ every transfer reaches a saver, so $dy = 0$: the limit $\frac{1}{1+\sigma\kappa}$ is not the value at $\chi = 1$.

Economic intuition. A transfer works only through people who spend it: borrowers at their limit spend every unit, a share $1 - \chi$ of the income this creates is spent again, and savers, pinned down by the real rate, ignore a transfer that their own future taxes repay. Who receives the transfer, not only its size, sets its effect.

Ref: Benigno (2015), §10, eqs. (25)–(32) and the fiscal remarks on p. 521; Romer (2012), §8.6 and §12.2.

Grading rubric.

- Keynesian cross from the budget constraint and m_T derived: 1.5 pts. m_T quoted: max. 0.5 pt. Transfer treated as a purchase g: max. half.
- Δ_e and its sign: 0.5 pt.
- Savers: 0.5 pt. Asserted without the Euler equation: 0 for this line.
- $\chi \to 1$ and the discontinuity: 0.5 pt.

d) (3 points) *Policy mix.* The bank follows the targeting rule of Question 4(a). Compare (i) cutting i subject to $i \geq 0$, (ii) transfers alone with $i = \rho$, and (iii) both. When the bound binds, find the transfer Δ_{zlb} that implements the optimum at $i = 0$ and the welfare cost of not paying it. Draw the AS–AD diagram: old and new AS; AD before the shocks, after them at $i = 0$, and after the policy; the equilibria.

Step — optimum — the rule $(y - y_e) + \theta(p - p^e) = 0$ and the new AS. $y^* = \frac{y_e + \theta\kappa y_n}{1+\theta\kappa} = y_n + \delta$ and $p^* - p^e = \kappa\delta$, with $\delta \equiv \frac{\mu}{(\sigma^{-1}+\eta)(1+\theta\kappa)} > 0$: the bank accommodates part of the markup shock.

Step — implementation — part (b) with $\zeta = \bar{y}_n - \varphi(i - \rho) + \frac{1-\chi}{\chi}(\hat{d} + \Delta)$. Policy moves (y, p) only through ζ, and $y = y^*$ iff $\zeta = y_n + (1 + \varpi\kappa)\delta$. The pairs (i, Δ) that implement the optimum form the line $\frac{1-\chi}{\chi}\Delta = \varphi(i - i^*)$, where

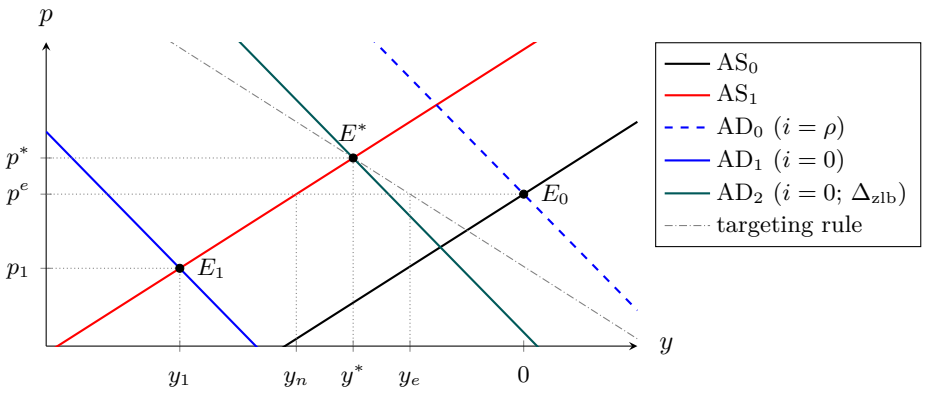
$$i^* = i_n - \frac{(1 + \varpi\kappa)\delta}{\varphi} < i_n.$$

Rate and transfer are perfect substitutes for (y, p); the bound binds for the optimum iff $i^* < 0$, a weaker condition than $i_n < 0$.

Step — the three options when $i^* < 0$. (i) At $i = 0$: $y_1 - y^* = \frac{\varphi i^*}{1+\varpi\kappa} < 0$ and $p_1 - p^* = \kappa(y_1 - y^*) < 0$. Along the AS the loss of Question 4(a) is $\frac{1}{2}(y - y_e)^2 + \frac{1}{2}\theta\kappa(y - y_n)^2$, a parabola in y with minimum at y^* and second derivative $1 + \theta\kappa$, so the bound costs $L_1 - L^* = \frac{(1+\theta\kappa)\varphi^2(i^*)^2}{2(1+\varpi\kappa)^2} > 0$. (ii) At $i = \rho$ the line gives $\Delta_\rho = \frac{\chi\varphi}{1-\chi}(\rho - i^*)$. (iii) At $i = 0$,

$$\Delta_{\text{zlb}} = \frac{\chi\varphi}{1-\chi}(-i^*) = \frac{\chi}{1-\chi}\left[(1 + \varpi\kappa)\delta - \varphi i_n\right] > 0,$$

the smallest transfer that attains L^* (on the line Δ rises with i), and $\Delta_\rho - \Delta_{\text{zlb}} = \frac{\chi\varphi\rho}{1-\chi} > 0$.



Economic intuition. Rate and transfer both move the AD intercept, so for (y, p) they are one instrument, and the transfer supplies what the bound withholds. If transfers cost anything outside the loss (public debt, later distortionary taxes), cut the rate first. As the optimum accommodates the markup shock, the bound binds for it before it binds for price stability.

Ref: Benigno (2015), §9; §11, eqs. (33)–(34); Romer (2012), §11.6.

Grading rubric.

- y^*, the implementing line and $i^* < i_n$: 1 pt.
- The three options, Δ_{zlb} and the welfare cost derived: 1 pt. Answered in the one-type model, without the borrowers' channel: max. half of (d).
- Graph (1 pt; four lines of 0.25 pt): axes and symbolic labels; AS_1 through (y_n, p^e); parallel AD shifts, AD_2 through E^*; E_1 below p^e, E^* on AS_1 and the rule.

e) (3 points) *Parameter analysis.* Sign, with proofs, the effects of χ, d_0 and κ (through α) on the transfer multiplier and on Δ_{zlb}, and collect them in a table; compare with the multiplier when inflation between the short and the long run is constant instead; give the limits and the stability conditions in both regimes; and read the results for Brazil's 2020 emergency transfers.

Step — multiplier — part (c). $\chi(1+\varpi\kappa) = \chi(1+\sigma\kappa) - \kappa d_0(1-\beta)(1-\chi)$, so $\frac{\partial m_T}{\partial \chi} = -m_T^2[1+\sigma\kappa+\kappa d_0(1-\beta)] < 0$, $\frac{\partial m_T}{\partial d_0} = m_T^2\kappa(1-\beta)(1-\chi) > 0$ and $\frac{\partial m_T}{\partial \kappa} = -m_T^2\chi\varpi$, negative iff $\varpi > 0$.

Step — needed transfer — part (d). $F \equiv (1-\chi)\Delta_{\text{zlb}} = \chi(1+\varpi\kappa)\delta - \chi\varphi i_n$, with $\chi\varphi i_n = \chi\varphi\rho + \chi(\bar{y}_n - y_n) + (1-\chi)\hat{d}$ and $\chi\varphi = \sigma\chi + (1-\chi)\beta d_0$; $y_n, \bar{y}_n$ depend on none of χ, d_0, α; δ on neither χ nor d_0 (it moves with α through κ). Hence $\frac{\partial \Delta_{\text{zlb}}}{\partial d_0} = -[\kappa(1-\beta)\delta + \beta\rho] < 0$ and, as $(1+\varpi\kappa)\delta \propto \frac{1+\varpi\kappa}{1+\theta\kappa}$, $\frac{\partial \Delta_{\text{zlb}}}{\partial \kappa} = \frac{\chi}{1-\chi} \cdot \frac{\mu(\varpi-\theta)}{(\sigma^{-1}+\eta)(1+\theta\kappa)^2}$. F is affine in χ, so $\frac{\partial \Delta_{\text{zlb}}}{\partial \chi} = \frac{F'(1-\chi)+F}{(1-\chi)^2} = \frac{F(1)}{(1-\chi)^2}$, with $F(1) = (1+\sigma\kappa)\delta - \sigma\rho - (\bar{y}_n - y_n) = -\sigma i^*_{\text{RA}}$, where i^*_{RA} is i^* at $\chi = 1$: the optimal rate of a one-type economy.

Step — constant inflation — $\bar{p}_n - p$ fixed in (32). Only the Fisher term moves with p: $dy = \frac{1-\chi}{\chi}\Delta + \frac{(1-\chi)d_0}{\chi}\kappa\, dy$, so $m_T^c = \frac{1}{\chi-(1-\chi)d_0\kappa}$, with $\frac{\partial m_T^c}{\partial \chi} = -(m_T^c)^2(1+d_0\kappa) < 0$, $\frac{\partial m_T^c}{\partial d_0} = (m_T^c)^2(1-\chi)\kappa > 0$, $\frac{\partial m_T^c}{\partial \kappa} = (m_T^c)^2(1-\chi)d_0 > 0$; and $\frac{1}{m_T} - \frac{1}{m_T^c} = \kappa\chi\varphi > 0$, so $m_T^c > m_T$.

	χ	d_0	κ (through α)
$m_T = 1/[\chi(1+\varpi\kappa)]$, anchored long run	$-$	$+$	$-$ iff $\varpi > 0$
$m_T^c = 1/[\chi-(1-\chi)d_0\kappa]$, constant inflation	$-$	$+$	$+$
Δ_{zlb}, transfer per borrower at $i = 0$	$+$ iff $i^*_{\text{RA}} < 0$	$-$	$-$ iff $\varpi < \theta$

Step — limits and stability. $\kappa \to 0$: $m_T, m_T^c \to \frac{1}{\chi}$ (the fixed-price cross). $\kappa \to \infty$ with $\varpi > 0$: $m_T \to 0$ (output stays at y_n). $d_0 \to 0$: $m_T \to \frac{1}{\chi(1+\sigma\kappa)}$, $m_T^c \to \frac{1}{\chi}$. $\chi \to 1$: $m_T \to \frac{1}{1+\sigma\kappa}$, $m_T^c \to 1$, and $\Delta_{\text{zlb}} \to +\infty$ if $i^*_{\text{RA}} < 0$. Stability (footnote 19: the AS flatter than an upward-sloping AD): anchored, $\kappa < -1/\varpi$ when $\varpi < 0$, i.e. $1+\varpi\kappa > 0$, automatic when $\varpi > 0$; constant inflation, $\kappa < \frac{\chi}{(1-\chi)d_0}$, i.e. $m_T^c > 0$. At each knife edge the multiplier diverges ($1+\varpi\kappa \to 0^+$, $\kappa \to \chi/[(1-\chi)d_0]^-$).

Step — reading. m_T falls with χ (fewer households re-spend income, and the real-rate brake ϖ rises toward σ), rises with d_0 (the inflation a transfer creates erodes more debt), and falls with κ when $\varpi > 0$ (flexible prices turn the push into inflation, which raises the real rate); under constant inflation the κ sign flips and the slump from $\hat{d}$, $m_T^c(1-\chi)\hat{d}$, deepens with κ (paradox of flexibility). Δ_{zlb} falls with d_0 (the cut to zero eases the borrowing limit more and m_T is larger) and with κ unless $\varpi > \theta$ (the rule accommodates less of the markup shock); it rises with χ when a one-type economy would itself need $i^* < 0$.

Step — Brazil, 2020. The *auxílio emergencial* paid lump sums to informal and low-income households, the closest analogue of Δ (households at their limit spend a transfer one for one). (1) Targeting sets the effect: a transfer spread over all households reaches borrowers only in proportion $1-\chi$; the part paid to savers does nothing. (2) Debt raises m_T and lowers the transfer needed

while long-run prices stay anchored; if inflation expectations de-anchor, $m_T^e > m_T$ but stability needs $\kappa < \chi/[(1-\chi)d_0]$. (3) The Selic stayed above zero, but an effective lower bound $\underline{i} > 0$ (risk premia, fiscal risk) works the same way: replace $-i^*$ by $\underline{i} - i^*$ in Δ_{zlb}. (4) After a cost-push shock the optimum has $p^* > p^e$: some inflation belongs to the optimal mix, and only a transfer above Δ_{zlb} moves the economy up AS_1 past E^*.

Economic intuition. Every sign weighs how much of a demand push survives the price response against how much of the economy re-spends income; anchoring of the long run decides whether price flexibility helps or hurts.

Ref: Benigno (2015), §10, footnotes 19–20 and the multipliers on p. 521; §11; Romer (2012), §8.6, §11.6, §12.2.

Grading rubric.

- Signs of m_T and m_T^e with their derivatives: 1 pt. Signs listed without derivatives: 0 for those lines.
- Signs of Δ_{zlb}, including the affine argument for χ: 0.75 pt.
- Limits and both stability conditions: 0.5 pt.
- Reading, including Brazil: 0.75 pt. Table without proofs or without reading: max. half of (e).